

MUHAMMAD AHMAD

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RESEARCH INTERESTS

Efficient machine learning: model compression and inference optimization for LLMs and VLMs, and hardware–software co-design under measured accelerator constraints.

EDUCATION

Lahore University of Management Sciences

Bachelor of Science in Electrical Engineering

Lahore, Pakistan

Aug. 2021 – Jun. 2025

- CGPA: 3.99/4.00 — Graduated with High Distinction
- **NMF Gold Medal and Fast Cables Award** — ranked first in the Electrical Engineering batch of 2025
- Dean's Honour List (2021–2024)
- Merit Scholarship (2021–2025)
- **Relevant coursework:** Advanced Topics in Machine Learning, Deep Learning, Foundations of Generative AI, Large Language Model Systems, Learning for Dynamics and Control, Distributed Systems, Data Engineering for AI Systems, Applied Probability

PUBLICATIONS AND WHITE PAPERS

QuantX: A Hardware-Aware Quantization Framework. Muhammad Ahmad, Khurram Mazher, Saqib Akram, Ahmad Tameem, Saad Bin Nasir.

Exploring the Application of Machine Learning for Soil Moisture Forecasting over In-situ Soil Moisture Sensor Networks. Muhammad Ahmad, Hamza Rafique, Abubakr Muhammad. *IEEE HITE*, 2024.

MANUSCRIPTS IN PREPARATION

Hardware–software co-design trade-offs of standard low-precision quantization formats for LLM/VLM inference. Analysis and profiling of the cost and compatibility of microscaling and conventional INT block quantization formats with existing outlier suppression techniques. *Target: MLSys 2027.*

Hardware-first quantization of Mixture-of-Experts models on non-GPU silicon. Kernel-profiled format selection and mixed-precision allocation for sub-35B MoE and non-MoE LLMs on the ET-SoC-1. *Target: ICLR 2027.*

RESEARCH EXPERIENCE

Hardware-Aware Compression and Inference Optimization

May 2025 – Present

Supervisors: Dr. Saad bin Nasir and Dr. Khurram Usman Mazher, 10xEngineers

Lahore, Pakistan

- **Bit-width dependence of rotation.** Benchmarked QuaRot and SpinQuant across the Llama-3 family under W4A4, W4A8, and W4A16 regimes, and extended the study to block rotations. Found rotation's benefit to be strongly bit-width dependent and neutral-to-detrimental at W4A8 and W4A16, contesting the uniform benefit assumed in prior work.
- **Failure modes under coarse scale rounding.** Showed that under E8M0 block scales (MX, BFP), the benefits of existing outlier suppression algorithms behave non-monotonically, due to the coarseness of scale. Traced the same coarseness to a failure of PEQA-style fine-tuning under E8M0, and showed that FP8-scaled formats such as NVFP4 remain tunable provided the inner scale range is bounded.
- **Custom quantization data formats design space.** Characterized viable combinations of group size, element dtype, scale dtype, outer scale, and asymmetric offset as well as failure modes. Showed max-invariant scale selection to be cheap but systematically suboptimal, and that importance-weighted (Hessian-diagonal) output-MSE grid search recovers most of the gap at a cost affordable for weights only.
- **Generalized scaling and clipping search.** Extended the AWQ scaling and clipping grid search from its default W4A16 setting to arbitrary $W_x A_y$ configurations; the generalized search matched or outperformed SmoothQuant with α -sweeps across every setting tested.
- **Hardware-first quantization of MoE models.** Profiled candidate numeric formats and matmul kernels for Qwen3.5-35B-A3B on the ET-SoC-1 with the compilers team, pruning the search space by measured throughput before accuracy tuning. Produced 4-bit and 3-bit recipes with simpler quantization formats attaining similar accuracy to existing open-source quants at higher speed when combined with outlier suppression.
- **Research mentorship.** Supervised two interns; one on diffusion-model quantization, the other on PEFT techniques for MXFP4 and NVFP4, covering literature direction, code review, and experiment design.

Undergraduate Thesis — Second-Order SAM for Federated Learning

Aug. 2024 – May 2025

Advisor: Dr. Muhammad Tahir, LUMS

Lahore, Pakistan

- Proposed a second-order perturbation formulation for SAM in federated settings: rather than the first-order gradient-ascent step, by aligning perturbation with the leading Hessian eigenvector obtained by power iteration, and average curvature regularized via a Hutchinson trace estimate.
- Evaluated in-distribution generalization of the proposed method on MNIST, CIFAR-10, and CIFAR-100 under Dirichlet label-skew partitions, and out-of-distribution transfer on PACS and OfficeHome, against FedAvg, FedSAM, and MoFedSAM baselines.

- Kept the method purely client-side, so that it composes with server-side aggregation fixes (gradient harmonization, control variates), which were implemented separately as comparison baselines.

Research Assistant — Neural Decoding of Cerebellar Activity

Jun. 2024 – Dec. 2024

Advisors: Dr. Muhammad Ali Siddiqi and Dr. Murtaza Taj, LUMS

Lahore, Pakistan

- Developed MLP and RNN-based decoders predicting lick behavior in mice from binary encoding of simple- and complex-spike activity, comparing feedforward and recurrent treatments of the temporal structure.
- Built upstream classifiers separating simple from complex-spike segments directly in raw neural recordings, allowing the decoding pipeline to run without hand-labeled spike-type annotation.

ONGOING RESEARCH

- Post-training quantization for Mixture-of-Experts architectures.** Router-weighted GPTQ, expert-aware scaling, clipping, and reordering search, and importance-weighted histogram construction for codebook quantization; mixed-precision allocation under kernel constraints that forbid per-expert precision assignment.
- Calibration-data composition for quantized reasoning models.** Effect of general-purpose corpora versus mathematical reasoning sets versus self-generated reasoning traces, and the sensitivity of reasoning and non-reasoning benchmarks to mixture ratio.
- Speed as a first-order objective in mixed-precision allocation.** Extending the MILP water-filling allocator to jointly optimize measured latency alongside accuracy sensitivity.
- KV-cache quantization** and PTQ for **GatedDeltaNet** and **Gated Attention** architectures.

WORK EXPERIENCE

Machine Learning Engineer

May 2025 – Present

10xEngineers

Lahore, Pakistan

- Lead developer of **QuantX**, a hardware-aware quantization framework for LLM and VLM inference (arXiv:2505.07531). Restructured it into a modular codebase, refactored the backend-core interfacing layer for the first release as a design-space explorer, and generalized the quantization primitive to arbitrary group size, element dtype, scale dtype, outer scale, and asymmetric offset — covering standard microscaling formats and custom quantization formats across the full outlier suppression and algorithm stack. Extended the model zoo to Qwen2–Qwen3.5 (including MoE), Qwen2.5-VL, Qwen3-VL, and SmolVLM.
- Wrote **MLIR kernels for 4-D block-quantized matmul** (`mmt4d_block`) in `shark-ai/sharktank`, extending per-channel support to blockwise quantization of both activations and weights, together with the Python dispatches and overloads; validated static and dynamic activation quantization and compiled through IREE with the compiler team for CPU and Milk-V Jupiter (RISC-V) targets.
- Built the **meta-optimizer**: an MILP water-filling mixed-precision allocator over layer sensitivity metrics with parameter-count weighting, replacing a hand-tuned, tensor-size-blind scheme. At ~ 4.6 average weight bits, it beat hand-configured 4-bit non-uniform codebook on **Llama-3.2-1B** (10.42 vs. 10.84 WikiText PPL; 11.41 for RTN) and on **Qwen3-VL-2B** (78.7 vs. 75.9 TextVQA, within 3% of BF16 at 4.3 bits/weight), while cutting quantization time from 30 to 7–25 GPU-minutes.
- Tuned BFP recipes for **Tenstorrent N150/N300** under native format constraints, where a column-major quantization axis leaves scales non-absorbable offline, and co-designed in-house codebook kernel support with the compiler team. On an N300 the deployable configuration lifted **Qwen-2.5-14B** top-1% accuracy from 89.6 to 94.4 over Tenstorrent’s published performance model, with a custom-dequant variant projected to hold 93.6 at lower DRAM (10.8 vs. 11.45 GB weight+KV).
- Benchmarked **Llama-3.1-8B** and **Qwen2.5-VL-7B** under BFP for a client deployment: meta-optimized configurations at ~ 4.5 average weight bits beat GPTQ+QuaRot on Llama-3.1-8B perplexity (7.51 vs. 7.68) and recovered Qwen2.5-VL-7B ChartQA to 76.1, against 76.2 for BF16 and 55.5 for round-to-nearest. Identified terminal Q and K activations as precision-critical outlier sites requiring high precision, online rotation, or post-RoPE smoothing.

TEACHING EXPERIENCE

Teaching Fellow — AI on Edge Devices (AI-624), Deep VLMs (AI-623)

Fall 2025 – Present

Lahore University of Management Sciences

Lahore, Pakistan

- AI-624, Fall 2026.** Assignment design and project supervision for the second offering, with projects running on Raspberry Pi 4 and 5 and Arduino Nano 33 BLE Sense alongside a track on compilers and kernel autotuning.
- AI-623, Spring 2026 (inaugural offering).** Designed and delivered the programming assignments, covering diffusion models (DDPM, DDIM), score-based models (NCSN), VQ-VAE, normalizing flows and flow matching, Langevin dynamics, and discrete diffusion; supported instruction on CLIP-style cross-modal alignment, early and late fusion, and SFT, LoRA, DPO, GRPO, and PPO. Assessed students by viva and programming evaluation.
- AI-624, Fall 2025 (inaugural offering).** Co-developed the assignment set for 50+ students covering knowledge distillation, PTQ and QAT, pruning, NAS, and efficient LLM deployment; ran labs on Raspberry Pi, Arduino Nano 33 BLE Sense, and TVM; mentored 10+ project groups on edge LLM inference with Ollama and llama.cpp and on static quantization for DiT-based diffusion models.

Teaching Assistant

Sep. 2023 – May 2025

Lahore University of Management Sciences

Lahore, Pakistan

- Supported six courses spanning circuits, applied probability, signals and systems, electromagnetic fields and waves, communication systems, and machine learning, reaching 50+ students.
- Ran tutorials and office hours, and authored supplementary problem sets and conceptual guides that were adopted into subsequent course offerings.

SELECTED PROJECTS

Data-Free Knowledge Distillation for Heterogeneous Federated Learning

Aug. 2024 – Dec. 2024

Advanced Topics in Machine Learning (CS-6304) course project

Lahore, Pakistan

- Replaced parameter averaging in federated aggregation with online multi-teacher, single-student distillation, using noise-engineered data-free distillation to avoid any data sharing and entropy- and diversity-based adaptive weighting to down-weight unreliable client teachers.
- Showed that purely data-free distillation is not competitive on its own, and proposed a serverless, client-only architecture in which a rotating designated client acts as the student and contributes its local data to the distillation step.
- Reached 70.3% top-1 on CIFAR-10 under Dirichlet($\beta=0.1$) label skew, against 43.6% for FedFTG and 43.3% for FedSAM; analyzed the residual convergence instability and its hyperparameter sensitivity.

Machine Learning for Earth Sciences

Jun. 2024 – Oct. 2024

WIT Summer Mentorship Program

Lahore, Pakistan

- Built an end-to-end pipeline forecasting soil-moisture signals from an IoT sensor network, using multivariate LSTMs for recursive N -day forecasting with autoregressive generation and feature selection by perturbation analysis.
- Confirmed predictive viability with RMSE and unbiased RMSD below standard thresholds and test-set correlation above 0.9; published as IEEE HITE 2024.

Modeling Water Management Dynamics in the Nile River Basin

Spring 2025

Modelling the Environment (ENV-244) course project

Lahore, Pakistan

- Built a Vensim system-dynamics model of the basin covering the Grand Ethiopian Renaissance Dam, the Aswan High Dam, and Jebel Aulia, calibrated against recorded inflows and reservoir capacities.
- Layered a second model adding domestic, agricultural, energy, and diplomatic pressures with feedback, exposed through a flight-simulator interface for exploring release fractions, withdrawal fractions, and capacity priority policies.
- Found agricultural and domestic demand growth to be the dominant driver of rising withdrawals, with diplomatic feedback partially offsetting downstream shortfall under coordinated release scenarios.

AI-LMS and ResearchExplorer

Mar. 2025 – Jun. 2025

LLM Systems and Data Engineering for AI Systems course projects

Lahore, Pakistan

- **AI-LMS:** a learning-management system with hierarchical course/topic/assignment storage on Supabase, generating and verifying assignments from PDFs and document URLs via a LangGraph and FastAPI RAG pipeline with rubric-based checking.
- **ResearchExplorer:** a Streamlit platform for research-trend analysis, pulling metadata, citations, and references from the Semantic Scholar API in batch and rendering citation graphs in Dgraph for citation-impact, co-authorship, and topic-evolution views.